

Python For Data Analytics

Total Time: 3 Hours

Theme: Python Foundations for Business Data Cleaning

Dataset: <https://www.kaggle.com/datasets/kyanyoga/sample-sales-data>

Section A: Concept Application

1. Identify the string methods and their execution order needed to remove extra spaces and fix inconsistent casing in a product name.
2. Explain why manual CSV loading often treats numeric columns as strings and identify the function used to convert them to floats.
3. Determine the best data structure for mapping product names to total sales and explain the drawback of using a list instead.
4. Compare using a manual for loop versus the `sum()` function, identifying a scenario where the loop is the only viable option.
5. Describe how to implement type checking or error handling to prevent a `TypeError` when comparing a potential string variable to a number.
6. Explain the benefits of modularizing a data pipeline into separate functions and the risks of using a single large code block in analytics.

Section B: Practical Task

1. Load the `sales_data_sample.csv` file and use Python string methods to trim whitespace and apply title casing to the 'PRODUCTLINE' column.
2. Iterate through the 'SALES' column to remove non-numeric characters and cast the cleaned values into a float data type.
3. Construct a dictionary where each key is a unique 'PRODUCTLINE' and the value is the cumulative sum of 'SALES' for that category.
4. Use a loop and conditional logic to categorize product lines as "High Revenue Product" if total sales exceed 100,000, printing the results to the console.

Section C: Mini Project

1. **Title:** Automated Sales Integrity & Reporting Engine
2. **Problem Statement:** Transform a fragmented regional retail export into a validated, high-integrity dataset by automating the identification of data type mismatches and revenue outliers.
3. **Dataset Recommendation:** Retail Sales Sample:
<https://www.kaggle.com/datasets/kyanyoga/sample-sales-data>
4. **Required Deliverables:** Python .ipynb script, automated console summary report, documented cleaning logic overview.